

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Technology Stack

ShopEZ E-Commerce Application

Phase: Requirement Analysis Phase

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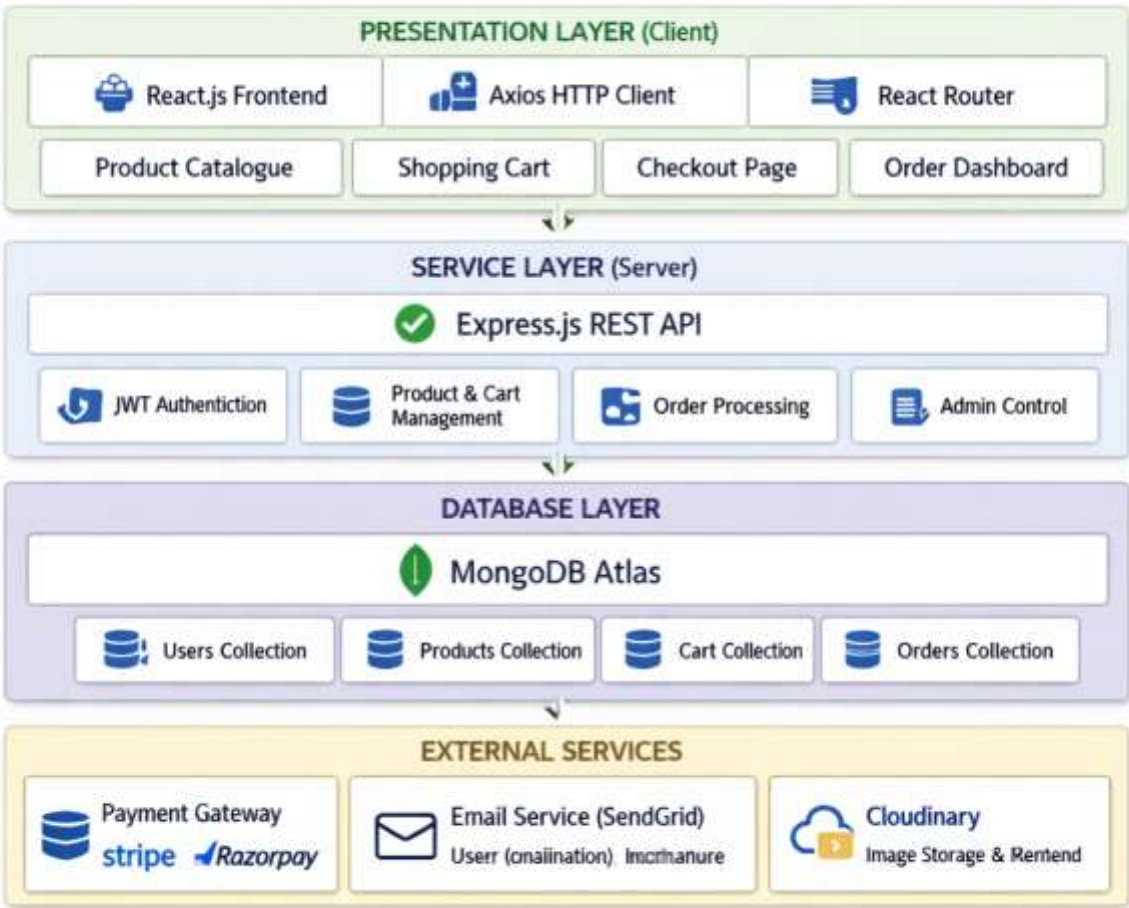
Classification: Project Documentation — For Academic Submission

Technology Stack

1. Introduction

This document specifies the **technology stack** adopted for the **ShopeZ E-Commerce Application**. Each technology is mapped to its architectural layer, purpose, and justification in supporting product browsing, cart management, checkout, payment processing, and order tracking functionalities.

2 ShopeZ E-Commerce Integration Architecture



3. Detailed Stack Breakdown

Layer	Technology	Version	Purpose & Justification
Frontend Framework	React.js	v18.x	Builds responsive, dynamic user interfaces for product browsing, cart, and checkout pages.
HTTP Client	Axios	v1.x	Manages communication between React frontend and Express backend APIs.
Backend Runtime	Node.js	v18 LTS	Executes server-side JavaScript and handles concurrent requests efficiently.
Backend Framework	Express.js	v4.x	Provides RESTful APIs for product, order, and user management..
Authentication	JSON Web Tokens (JWT)	Latest	Secures user sessions and protects private routes for customers and admins.

Layer	Technology	Version	Purpose & Justification
Password Security	Bcrypt.js	v5.x	Encrypts user passwords before storing them in the database.
Database	MongoDB (Atlas)	v6.x	Stores user, product, cart, and order data in a scalable cloud database.
ODM	Mongoose	v8.x	Simplifies MongoDB operations and enforces schema validation.
Payment Integration	Razorpay / Stripe API	Latest	Enables secure online payment processing during checkout.
Deployment	Vercel (FE) / Render (BE)	Latest	Hosts the application online with seamless CI/CD and maintenance..

4. Architecture Decision Records (ADRs)

ADR-01: MongoDB over Relational Database

Field	Detail
Status	✓ Accepted
Decision	Use MongoDB as the primary database.
Justification	Flexible schema supports product catalogs, orders, and user data efficiently..
Consequences	Requires schema validation through Mongoose models.

ADR-02: JWT Authentication over Session-Based Authentication

Field	Detail
Status	✓ Accepted
Decision	Use JWT tokens for authentication.
Justification	Stateless authentication improves scalability and API security.
Consequences	Token expiration and secure storage must be handled properly.

ADR-03:ReactFrontend over Traditional Server-Side Rendering

Field	Detail
Status	☒ Accepted
Decision	Bulid frontend using React.js.
Justification	Provides a fast and interactive user experience for shopping and order tracking.
Consequences	Requires API integration with the backend.